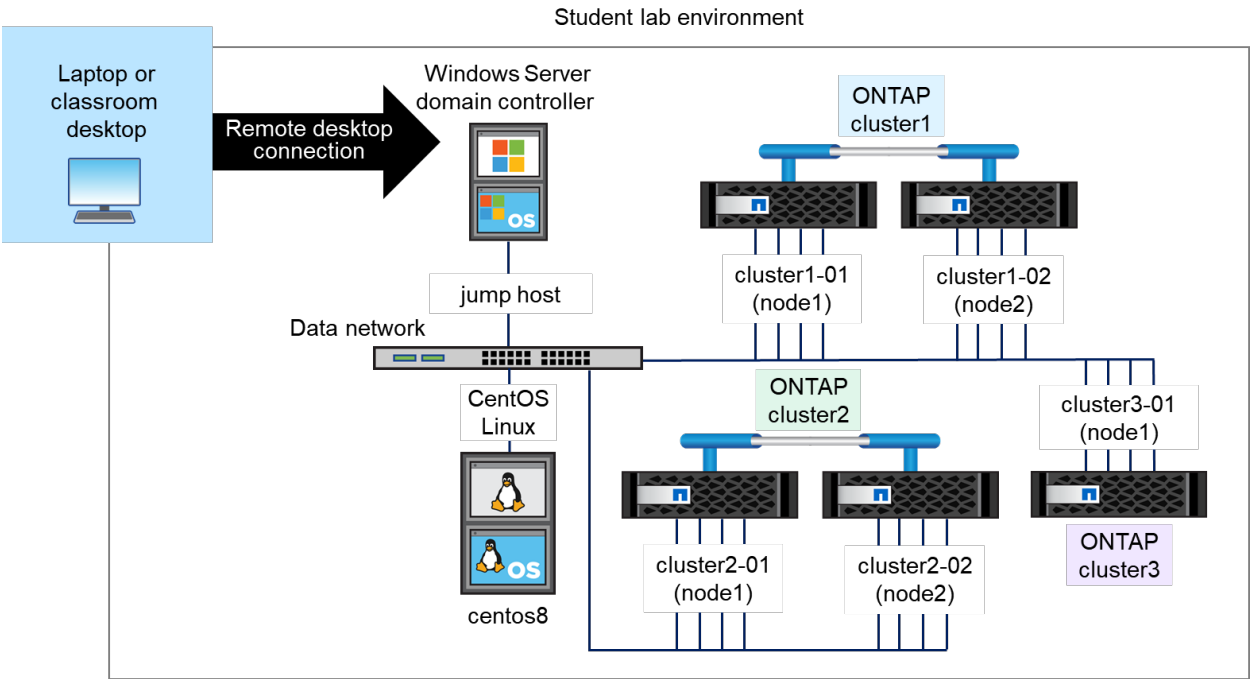


Module 3: SnapMirror Synchronous for Disaster Recovery

Exercise Equipment Diagram



System	Host Name	IP Addresses	User Name	Password
Windows Server 2022 Datacenter	JUMPHOST	192.168.0.5	DEMO\Administrator	Netapp1!
ONTAP cluster-management LIF (Cluster1)	Cluster1	192.168.0.101	admin (case-sensitive)	Netapp1!
node 1 (Cluster1)	cluster1-01-console	192.168.0.111	admin (case-sensitive)	Netapp1!
node 2 (Cluster1)	cluster1-02-console	192.168.0.112	admin (case-sensitive)	Netapp1!
ONTAP cluster-management LIF (Cluster2)	Cluster2	192.168.0.102	admin (case-sensitive)	Netapp1!
node 1 (Cluster2)	cluster2-01-console	192.168.0.113	admin (case-sensitive)	Netapp1!
node 2 (Cluster2)	cluster2-02-console	192.168.0.114	admin (case-sensitive)	Netapp1!
ONTAP cluster-management LIF (Cluster3)	Cluster3	192.168.0.103	admin (case-sensitive)	Netapp1!
node 1 (Cluster3)	cluster3-01-console	192.168.0.115	admin (case-sensitive)	Netapp1!
Linux Server	centos8	192.168.0.21	root	Netapp1!

Exercise 1: Configuring a SnapMirror Synchronous Relationship

In this exercise, you configure a SnapMirror synchronous (SM-S) relationship between two volumes.

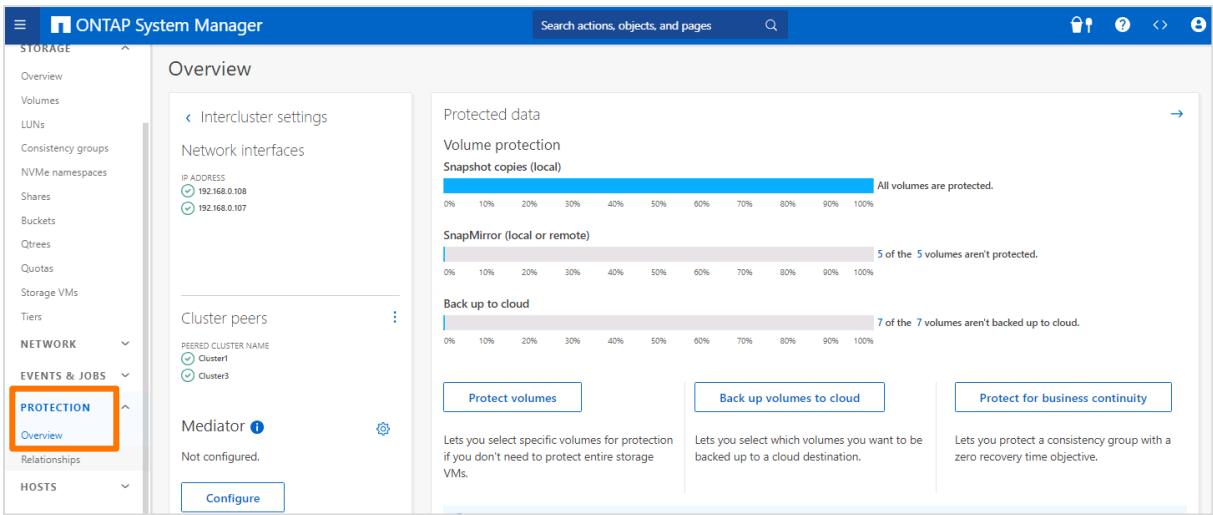
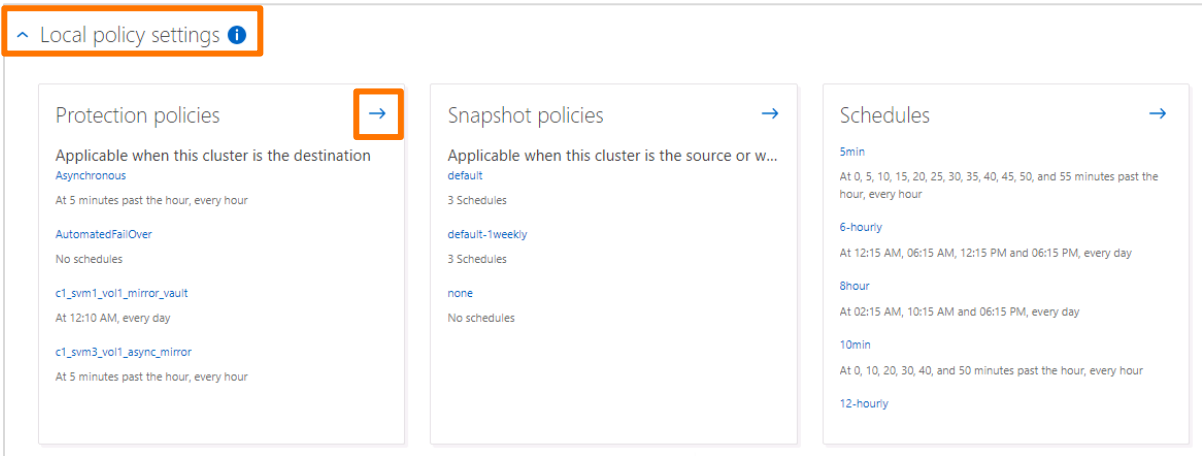
Objectives

This exercise focuses on enabling you to do the following:

- Create an SM-S replication policy
- Create an SM-S relationship

Task 1: Create an SM-S Policy

In this task, you create an SM-S policy and relationship.

Step	Action
1-1	In the System Manager navigation pane for Cluster2, select Protection > Overview. 
1-2	In the Overview pane, expand Local policy settings, and then click the arrow in the upper-right corner of Protection policies. 

Step	Action																		
1-3	On the Protection policies tab, click Add to create a policy. Policies Protection overview Protection policies Snapshot policies + Add <table><thead><tr><th>Name</th><th>Description</th><th>Policy type</th></tr></thead><tbody><tr><td>Asynchronous</td><td>A unified Asynchronous SnapMirror and SnapVault policy for mirroring the latest ...</td><td>Asynchronous</td></tr><tr><td>AutomatedFailOver</td><td>Policy for SnapMirror Synchronous with zero RTO guarantee where client I/O will ...</td><td>Synchronous</td></tr><tr><td>c1_svm1_vol1_mirror_vault</td><td>cluster1:c1_svm1_vol1 – Unified SM Policy</td><td>Asynchronous</td></tr><tr><td>c1_svm3_vol1_async_mirror</td><td>cluster1:svm3_vol1 – Async Mirror policy</td><td>Asynchronous</td></tr><tr><td>CloudBackupDefault</td><td>Vault policy with daily rule.</td><td>Asynchronous</td></tr></tbody></table>	Name	Description	Policy type	Asynchronous	A unified Asynchronous SnapMirror and SnapVault policy for mirroring the latest ...	Asynchronous	AutomatedFailOver	Policy for SnapMirror Synchronous with zero RTO guarantee where client I/O will ...	Synchronous	c1_svm1_vol1_mirror_vault	cluster1:c1_svm1_vol1 – Unified SM Policy	Asynchronous	c1_svm3_vol1_async_mirror	cluster1:svm3_vol1 – Async Mirror policy	Asynchronous	CloudBackupDefault	Vault policy with daily rule.	Asynchronous
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CloudBackupDefault	Vault policy with daily rule.	Asynchronous																	
1-4	 In the following step, you select the Synchronous policy type before entering the policy name. If you select the Synchronous policy type after entering the policy name, the policy name changes to a default value. You then need to reenter the policy name.																		

1-5

In the Add protection policy dialog box, specify the following values:

- Policy type: **Synchronous**
- Policy Name: **c1_svm2_vol1_synchronous_mirror**
- Policy Description: **SM-S Mode**
- Policy Scope: **Cluster**
- Continue data services even if the copy can't be synchronized: *Select checkbox*
- Allow simultaneous access to the destination copy: *Clear checkbox*

Add protection policy

POLICY NAME

POLICY DESCRIPTION

POLICY SCOPE

☒ Cluster
 ☐ Storage VM

Policy type

☐ Asynchronous
The copy can be used for backup and disaster protection.
 ☒ Synchronous
The copy matches the source.
 ☐ Continuous
Continuously back up to on-premises or cloud object store.

☒ Continue data services even if the copy can't be synchronized.
 ☐ Allow simultaneous access to the destination copy.

Rules

Rule considerations

Matching SnapMirror label	Retention count
No data	

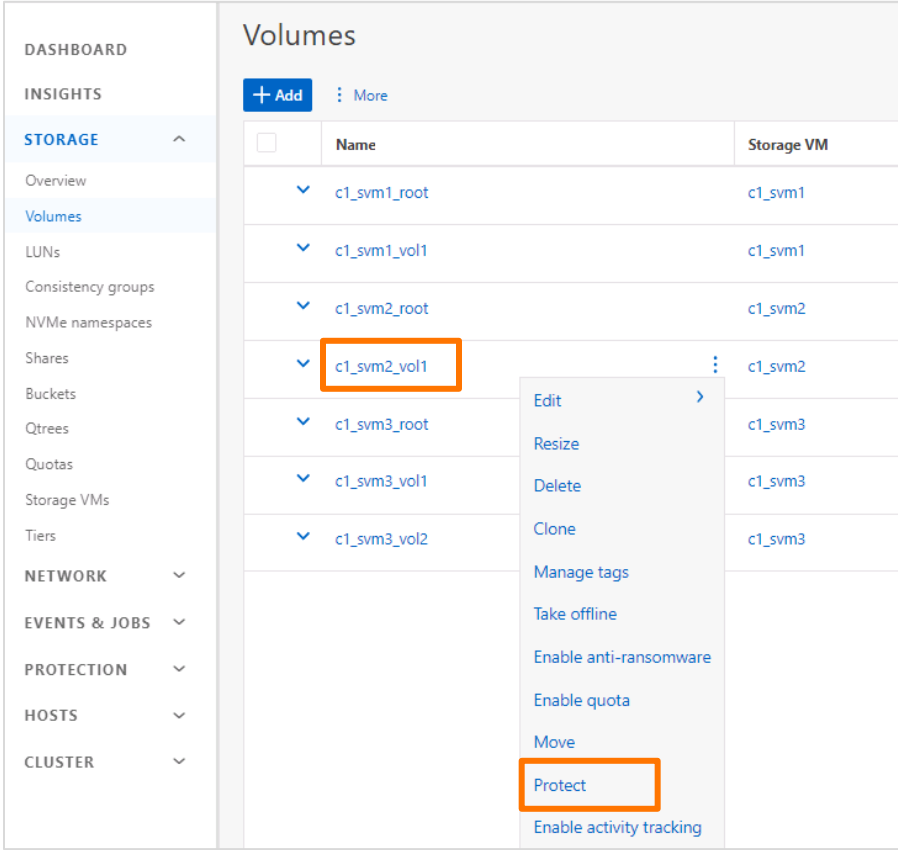
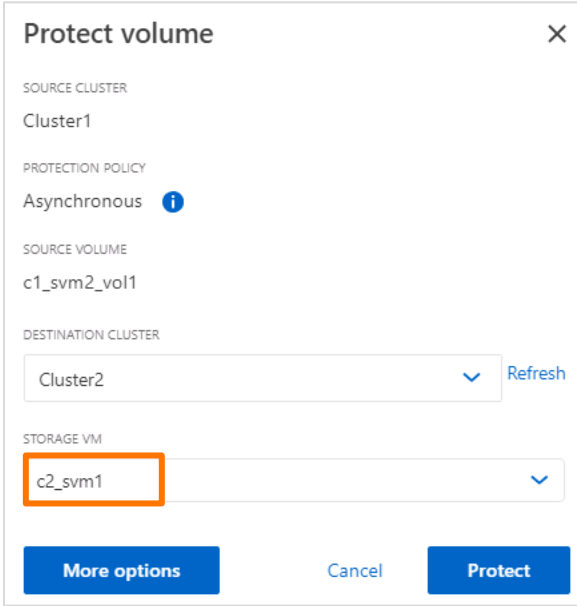
+ Add

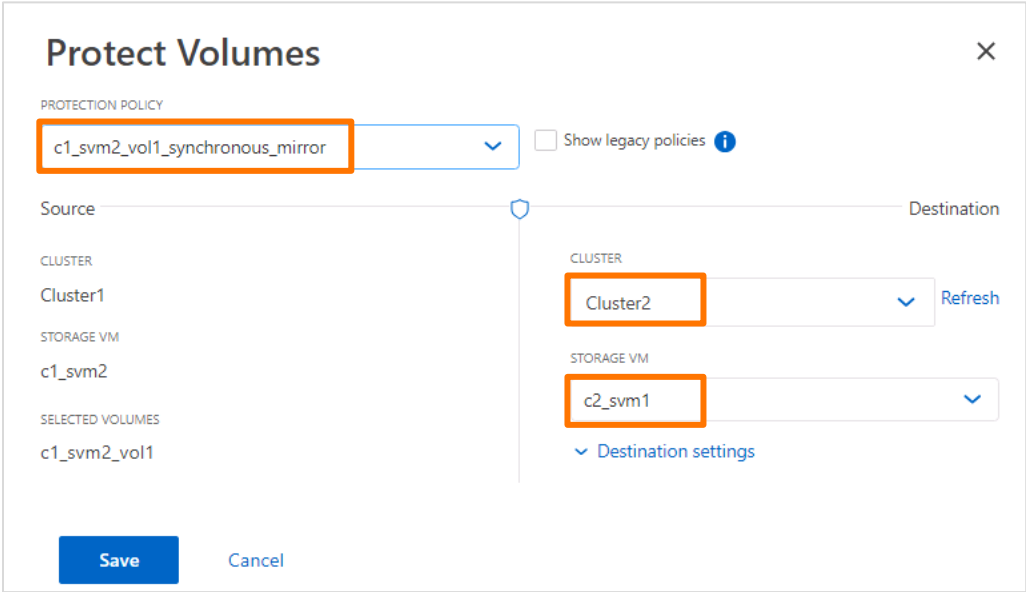
Save

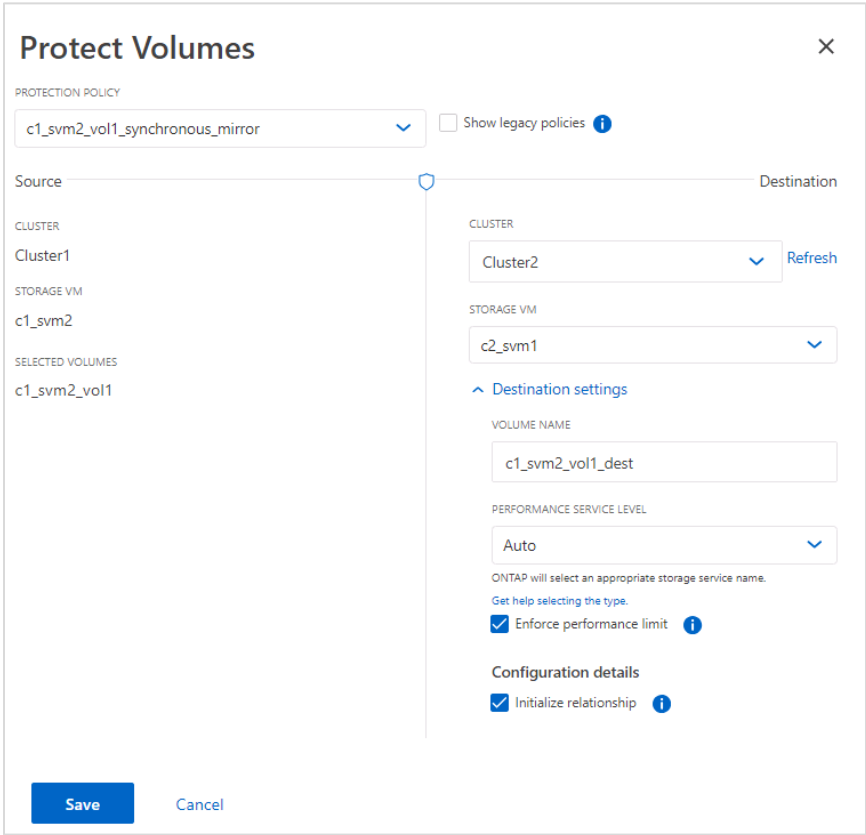
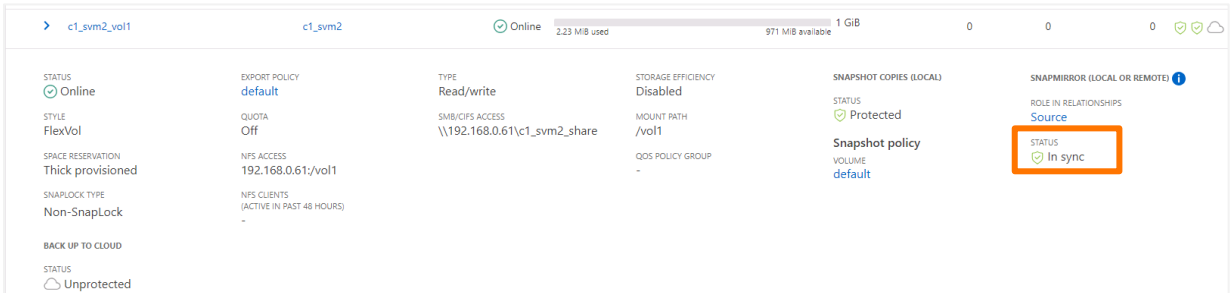
Cancel

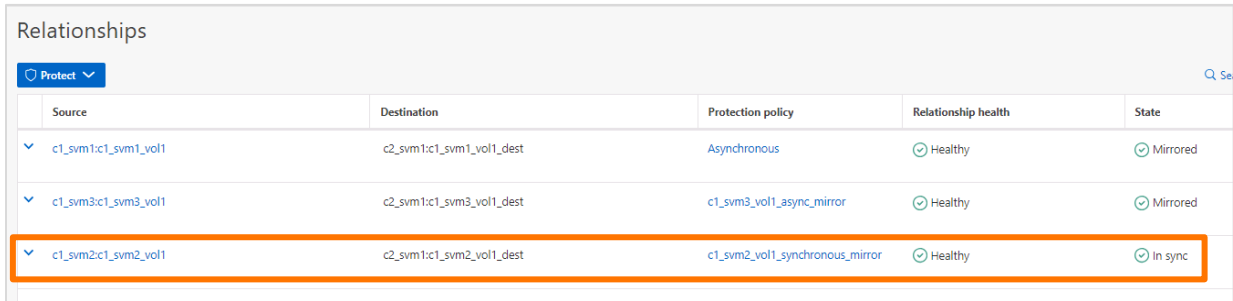
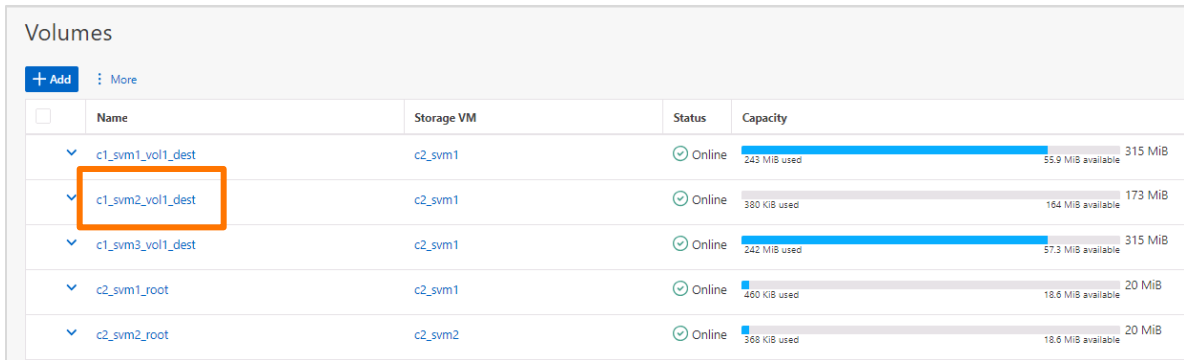
Step	Action																								
	Add protection policy Policy name sync_policy_861 Policy description Policy scope ☒ Cluster ☐ Storage VM Policy type ☐ Asynchronous The copy can be used for backup and disaster protection. ☒ Synchronous The copy matches the source. ☐ Continuous Continuously back up to on-premises or cloud object store. ☐ Continue data services even if the copy can't be synchronized. ☐ Serve I/O only from the primary copy. ☐ Serve I/O from both copies. Rules Rule considerations Matching SnapMirror label Retention count SnapLock retention period No rules + Add Save Cancel																								
1-6	Click Save .																								
1-7	 Synchronous policy is enabled by selecting Synchronous for the policy type and selecting the Continue data services even if the copy can't be synchronized checkbox.																								
1-8	On the Protection policies tab, verify that the policy was created successfully. Policies Protection overview Protection policies Snapshot policies + Add <table><tr><th>Name</th><th>Description</th><th>Policy type</th></tr><tr><td>Asynchronous</td><td>A unified Asynchronous SnapMirror and SnapVault policy for mirroring the latest ...</td><td>Asynchronous</td></tr><tr><td>AutomatedFailOver</td><td>Policy for SnapMirror Synchronous with zero RTO guarantee where client I/O will ...</td><td>Synchronous</td></tr><tr><td>c1_svm1_vol1_mirror_vault</td><td>cluster1:c1_svm1_vol1 – Unified SM Policy</td><td>Asynchronous</td></tr><tr><td>c1_svm2_vol1_synchronous_mirror</td><td>SM-S Mode</td><td>Synchronous</td></tr><tr><td>c1_svm3_vol1_async_mirror</td><td>cluster1:svm3_vol1 – Async Mirror policy</td><td>Asynchronous</td></tr><tr><td>CloudBackupDefault</td><td>Vault policy with daily rule.</td><td>Asynchronous</td></tr><tr><td>Continuous</td><td>Policy for S3 bucket mirroring.</td><td>Continuous</td></tr></table>	Name	Description	Policy type	Asynchronous	A unified Asynchronous SnapMirror and SnapVault policy for mirroring the latest ...	Asynchronous	AutomatedFailOver	Policy for SnapMirror Synchronous with zero RTO guarantee where client I/O will ...	Synchronous	c1_svm1_vol1_mirror_vault	cluster1:c1_svm1_vol1 – Unified SM Policy	Asynchronous	c1_svm2_vol1_synchronous_mirror	SM-S Mode	Synchronous	c1_svm3_vol1_async_mirror	cluster1:svm3_vol1 – Async Mirror policy	Asynchronous	CloudBackupDefault	Vault policy with daily rule.	Asynchronous	Continuous	Policy for S3 bucket mirroring.	Continuous
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Continuous	Policy for S3 bucket mirroring.	Continuous																							

Task 2: Create an SM-S Relationship

Step	Action
2-1	In the System Manager navigation pane for Cluster1, select Storage > Volumes .
2-2	Click the three vertical dots to the right of c1_svm2_vol1 and then select Protect . 
2-3	In the Protect volume dialog box, select c2_svm1 for the destination SVM. 

Step	Action
2-4	Click More options .
2-5	In the Protect Volumes pane, specify the following values: - Protection Policy: c1_svm2_vol1_synchronous_mirror - Destination > Cluster: Cluster2 - Destination > Storage VM: c2_svm1 

Step	Action
2-6	Expand Destination settings and then review the values that are listed without making any changes. 
2-7	Click Save .
2-8	 Cluster1 and Cluster2 are currently in a peer relationship. On Cluster1, c1_svm2 is not in a peer relationship with c2_svm1 on Cluster2. However, after you click Save in the previous step, System Manager automatically peers the two SVMs and then creates the SnapMirror relationship.
2-9	 The Adding the relationship status bar appears for a few seconds. Optionally, click Run in Background.
2-10	In the Volumes pane, click the down arrow next to c1_svm2_vol1 to view the details and verify that the protection relationship was created successfully, and see that the status is In sync. 

Step	Action
2-11	In the System Manager navigation pane for Cluster2, select Protection > Relationships .
2-12	In the Relationships pane, verify that the SM-S relationship is listed and that the Relationship health is Healthy. 
2-13	In the navigation pane, select Storage > Volumes .
2-14	In the Volumes pane, verify that the destination volume for the SM-S relationship is listed. 
2-15	 Optionally, you can map a drive to the source volume and copy File_240M from C:\CourseFiles to the mapped drive. After the data transfer, you can review the storage capacity that is used by the destination volume.

End of exercise